System Overview

***Note**: This is background information for the entire platform, not just the database.*

THS wants to develop and adopt a new social media platform exclusively for university students, referred to as **KTH Social**. This platform is aimed at making it easier for university students and staff members to connect with each other, schedule events and share posts. Let's take a look at the main use cases for this platform:

Posts: The platform allows users to share their thoughts and ideas through posts containing text, image or video. These posts may be sorted based on place or tags. Furthermore, in order to promote high quality content, users can like other posts.

Event management: The platform lets users create events with a place and a date. Other users can sign up to attend these events.

Connecting with others: The platform lets users connect with each other through friendships. This lets users keep track of their friends' posts and the events they are arranging or attending.

Subscription: Through monthly subscriptions users are able to gain access to premium features such as longer texts etc.

Database Data Requirements

***Important**: The relations/tables you create do not have to follow the structure used in the following list.*

1. Required information about **students**:
    - User ID
    - Full Name
    - Programme
    - Enrollment year
    - Friends (by their User IDs)
2. Required information about **staff**:
    - User ID
    - Full Name
    - Department
    - Friends (by their User IDs)
3. Required Information about **posts:**
    - Post URL
    - User ID
    - Title
    - Timestamp
    - Place[1]
    - Tags
    - Body text
4. Required information about **likes**:
    - Post URL
    - User ID
    - Timestamp
5. Required information about **attachments:**
    - Post URL

[1] Address of a place related to the post.

*The following document represents a study case for the database technology course DD1368, and a source document to be applied in the course exercises including homeworks, labs and discussion sessions.

- Attachment index
- File extension
- File size

6. Required information about **events**:
    - Event ID
    - User ID
    - Title
    - Place
    - Start Time
    - End Time
    - Duration
    - Attendees
7. Required information about **subscription**:
    - Subscription ID
    - Date of Payment
    - User ID
    - Payment Method
    - Expiry date

Important notes:

- A Post URL, User ID, Event ID or Subscription ID may only refer to one single post, user, event or subscription, respectively.
- Friends is a symmetric relationship, meaning that if A is friends with B, then B must also be friends with A.
- The User ID in Event refers to the user who arranges the event.
- Out of the listed attributes, only the following can have null data instances in the database:

  - Tags
  - Place of Post
  - Title of Post
  - Body text of Post

Detailed Requirements

Note: These requirements might need new attributes or even new tables besides the ones already created based on the Data Requirements. A good indicator that the database has been designed correctly is that these requirements can be checked fairly easily using queries without subqueries.

A. A user can create their own tags and a post can have any number of tags that the user specifies.
B. A user can be both a student and staff.
C. Both staff and students can post and create events.
D. A student can only study one programme.
E. The departments that a staff can be a part of are: {ABE, EECS, ITM, CBH, SCI}
F. A user can only have one active subscription at a time.
G. A user should be able to view all of their expired subscriptions.
H. A subscription is paid monthly and is valid for 30 days after the transaction is made. It can not be terminated early.
I. A subscription can be paid either using Klarna, Swish, Card or Bitcoin.

J. A place is represented by an address that the user may specify.
K. The person who arranges an event may also attend that event.
L. The attachment index begins at 1 for each post and every additional attachment in a post will increment this index by 1
M. An attachment is accessed by a URL that is formatted as a combination of the postURL and fields in attachment: postURL + “/” + attachmentIndex + “.” + fileExtension. E.g http://www.kthsocial/1fa51/3.jpg
N. Allowed file type extensions are jpg (images), mp4 (videos), m4a (audios), pdf (documents)
O. Timestamps are stored in the format YYYY-MM-DD HH:MM:SS.
P. Users are limited to one post per second
Q. Only users of KTH-Social can attend events on the platform